2025 10th International Conference on Intelligent Computing and Signal Processing (ICSP) | 979-8-3315-3626-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSP65755.2025.11086925

SLM-based Hybrid Retrieval for Resource Constrained Retrieval-Augmented Generation on Open Super-Large Crawled Data

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***Abstract*—Large Language Models (LLMs) have significantly advanced the accessibility of natural language processing across diverse applications. However, susceptibility to hallucinations and reliance on outdated data limit their reliability. Retrieval-Augmented Generation (RAG) systems address this by providing access to external, up-to-date information during inference. Despite this improvement, existing dense, sparse, and hybrid retrieval strategies often struggle to balance semantic relevance and efficiency. In this study, we implement a hybrid retrieval approach that utilizes a small language model (SLM)-based reranking. We compare its performance against traditional hybrid retrieval methods combining FAISS HNSW with BM25 and TF-IDF reranking on a 500,000-document subset from OSCAR, evaluated under realistic compute limits using a single-GPU Colab environment (L4 GPU, 22.5 GB VRAM) to simulate resource-constrained RAG settings. We evaluate retrieval quality (precision, Recall) and efficiency (seconds-per-query, query-per-seconds). Results show that hybrid BM25 (0.05 SPQ, 21.96 QPS) and TF-IDF (0.01 SPQ, 71.17 QPS) methods are more efficient, but the SLM-based hybrid achieves superior retrieval accuracy. It substantially outperforms both traditional approaches in top-5 relevance, achieving a Precision@5 of 0.028 and a Recall of 0.102.**

Keywords—Generative AI; Big Data; Information Retrieval

I. Introduction

The swift progress of large language models (LLMs) has enhanced the convenience and accessibility of natural language processing [1]. These have empowered applications with valuable features like chatbots and domain-specific QA systems. However, because hallucinations and stale knowledge limit their real-world use, researchers have turned to Retrieval-Augmented Generation (RAG) as a practical remedy [2]. RAG systems enhance LLMs by retrieving up-to-date information from an external data source at inference time. However, their deployment on devices or services with limited memory, computing, and strict latency budgets—so-called resource-constrained environments [3]—poses significant challenges when scaling to open-domain tasks. Recent work has focused on retrieval strategies that balance both computationally efficient [4] and the capability to handle massive corpora [5].

RAG retrieval methods fall into two main classes—dense and sparse—with complementary strengths and weaknesses. Neither approach alone meets both speed and semantic coverage needs at scale. Hybrid retrieval strategies combine dense and sparse retrieval techniques to solve these limitations.

This paper implements an SLM-based hybrid retrieval algorithm for low-latency operation in resource-constrained settings. The algorithm leverages a small language model (SLM) for automatic document relevance checking. We then evaluate the algorithm comprehensively and compare it to classical and hybrid retrieval algorithms over extensive open crawled data. Unlike traditional RAG pipelines deployed on high-throughput clusters, this study simulates realistic deployment conditions where hardware availability is limited to a single GPU instance.

The rest of the paper is organized as follows: Section II examines similar work, Section III describes the methodology, Section IV discusses the results, and Section V concludes with recommendations.

II. Related Work

Retrieval algorithms are essential in RAG systems because they enhance domain-specific expertise, facilitate timely updates without retraining, and increase the interpretability and controllability of model outputs by aligning them closely with factual data. This section explores existing work with Retrieval algorithms.

BM25 (also known as Best Match 25 or Okapi BM25) is a sparse retriever and one of the most widely used classical information retrieval techniques. Due to its interpretable nature and applicability in diverse applications, BM25 is a simple and computationally efficient algorithm [6]. The downside to BM25 is that it excels in capturing semantic meaning within exact keyword matches.

Term Frequency-Inverse Document Frequency (TF-IDF) is an established statistical approach in information retrieval. This technique for ranking documents based on relevance aims to capture the significance of words within a corpus by evaluating terms according to their frequency and inverse document frequency [6]. TF-IDF also struggles to capture contextual nuances [7], being unable to detect synonyms and related terms, possibly leading to the retrieval of irrelevant documents or incomplete information [6].

FAISS (Facebook AI Similarity Search) is an open-source tool developed by Meta AI Research. It has significant applications and is well-known in large-scale data fields,

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including visual and text retrieval, especially natural language processing tasks. FAISS integrates Hierarchical Navigable Small World (HNSW) as one of its indexing methods for approximate nearest neighbor search due to the common use of high-dimensional vector representations in RAG systems. Its strength is its remarkable scalability, enabling rapid search even in extremely large-scale datasets [6]. However, despite these advantages, it needs much memory to store its complex hierarchical graph structures.

Small Language Models (SLMs) have also been a field of research in hybrid Retrieval for RAG systems. Recent studies, such as MiniRAG [9] and RoseRAG [10], have begun exploring lightweight architectures that leverage SLMs for efficient retrieval and generation in resource-constrained environments. These efforts primarily focus on architectural simplification, distributed inference, or margin-based optimization. Additionally, few works have systematically examined the performance trade-offs between SLM-based Hybrid reranking and traditional sparse or dense retrievers under strict computational limitations.

III. Methodology

A. Datasets

This paper utilizes two datasets: the Open Super-large Crawled ALMAnaCH coRpus (OSCAR) and FinQA datasets.

- This study utilizes the OSCAR dataset due to its large size. Its English language distribution is 3.2 TB, with exactly 431,992,659 documents and 377,376,402,775 words. The retriever will go over 500,000 samples.
- The FinQA dataset contains 7,000 question-answer pairs derived from company financial reports, specifically the 10-K filings.

The OSCAR dataset represents an open, super-large, and linguistically diverse corpus. Even at the 500,000-document subset level, its scale demands efficient indexing and scalable retrieval strategies, making it a suitable benchmark for testing hybrid methods in RAG systems.

B. Computing Environment

For development and all testing, a single Google Colab environment equipped with 53.0 GB System RAM and an L4 GPU with 22.5 GB GPU RAM was used. While sufficient for model prototyping, this setup contrasts with typical RAG deployments on distributed or multi-GPU clusters. Its limitations in memory bandwidth and concurrent model execution highlight the feasibility of deploying hybrid retrieval in constrained environments.

C. Methodological Framework

Figure 1 outlines the methodological framework used in this study. Five hundred thousand samples from the OSCAR dataset were preprocessed by chunking the data into fixed-size segments–the 500,000-document subset reflects a scalable segment of OSCAR's complete 3.2TB corpus, offering a balance between volume and feasibility for single-GPU experimentation. These were then passed through the all-MiniLM-L6-v2 sentence transformer to generate embeddings. Stored in a single structure and indexed using the FAISS HNSW algorithm for approximate nearest neighbor retrieval, with parameters selected based on best-practice tuning guidelines for FAISS HNSW [11], balancing search depth and index construction time: M = 64, efConstruction = 200, and efSearch = 128.

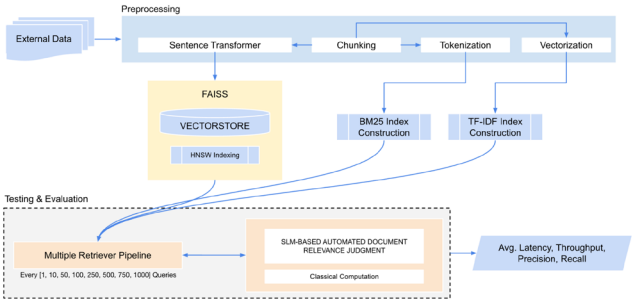


Fig. 1 Methodological Framework

The same chunked data was used to construct the BM25 and TF-IDF indexes. For BM25, an in-memory inverted index is created: after receiving the tokenized chunk, it computes each term's IDF once and stores term frequencies. The result is a large, sparse matrix that quickly scores how closely any query matches each document. Once indexing is finished, the classical and hybrid retrievers are built.

Each retriever received the same 7,000 queries. Time values were recorded for every 1, 10, 50, 100, 250, 500, 750, and 1000 queries to compute the average latency and throughput.

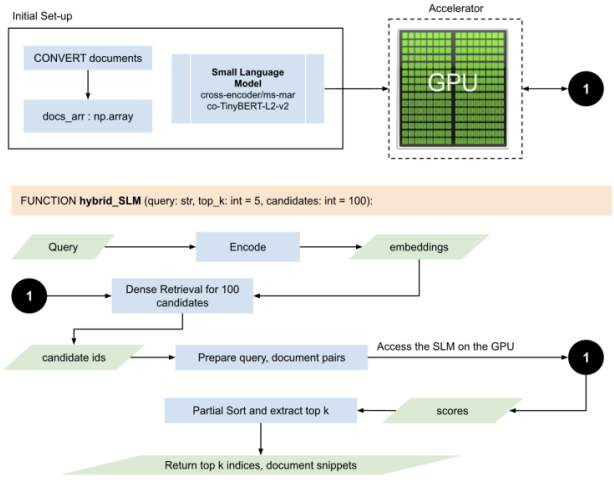


Fig. 2 Schematic of the two-phase SLM-based Hybrid Retrieval process, including FAISS candidate selection and SLM reranking.

Precision and Recall values were computed using the SLM-based document relevance judgment, wherein all 7,000 question fields in the FinQA dataset were used as queries.

The SLM-based Hybrid Retriever employs a two-phase retrieval process entirely on the GPU. In the first phase, it encodes the user's query using a sentence embedding model. It queries a FAISS HNSW index to retrieve a fixed number of candidate documents based on vector similarity. In the second phase, it reranks these candidates using a cross-encoder model that scores each (query, document) pair for relevance. The top-k highest-scoring documents are then selected and returned. This approach combines fast, dense retrieval with accurate

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neural reranking, optimizing speed and precision without requiring CPU-GPU data transfers.

D. Performance Metrics

The retrieval algorithms' performance was evaluated based on the following metrics: average latency, throughput, and SLM-based Document Relevance Judgment for Precision and Recall.

- Average latency (seconds-per-query / SPQ) quantifies the duration the retrieval system requires to obtain pertinent documents in reply to a query.
- Throughput (queries-per-second / QPS) indicates the number of retrieval tasks the system can execute in a designated time period, demonstrating its ability to manage large quantities of queries.
- Precision evaluates the ratio of retrieved documents that are pertinent to the query.
- Recall determines the number of relevant documents that were successfully obtained.

IV. Results & Comparative Analysis

The three hybrid algorithms were evaluated on 7,000 FinQA queries. Relevance judgments were performed using an SLM-based cross-encoder, specifically mxbai-rerank-xsmall-v1 by Mixedbread. Table I summarizes the initial system performance across five retrieval strategies using 10 FinQA queries.

TABLE I. Summary of Performance with 10 Queries

Method	Total Time (s)	Avg Latency (SPQ)	Throughput (QPS)
BM25	36.71	3.67	0.27
TF–IDF	8.03	0.80	1.25
Hybrid BM25	0.46	0.05	21.96
Hybrid TF-IDF	0.14	0.01	71.17
SLM-based Hybrid	2.05	0.20	4.88

Among the evaluated methods, classical BM25 exhibited the longest total execution time, highest latency, and lowest throughput. Similarly, classical TF-IDF demonstrated inferior performance compared to all hybrid approaches. BM25 operates on exact lexical matches and lacks semantic understanding. It typically executes on the CPU, resulting in higher latency and lower throughput, particularly in large-scale retrieval tasks. On the other hand, TF-IDF weights terms by frequency but, like BM25, does not capture semantic similarity. Although faster than BM25, it remains limited in ranking relevance and outperforms hybrid and embedding-based methods. The hybrid retrieval methods significantly outperformed the classical baselines across all efficiency metrics.

In Figures 3 and 4, we examine the performance of the hybrid retrieval algorithms by executing a series of FinQA queries at increasing scales: 1, 10, 50, 100, 250, 500, 750, and 1000. This progression enables a more granular analysis of each method's behavior under different workloads.

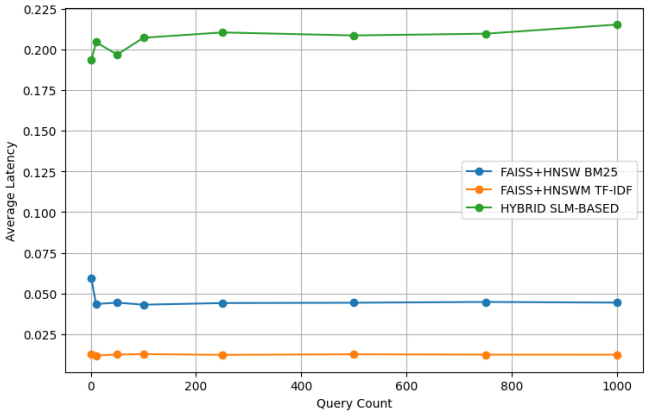


Fig. 3 Average Latency after 1,000 FinQA Queries

As illustrated in Figure 3, although all methods leverage a FAISS index for candidate filtering, the SLM-based hybrid introduces a second, GPU-intensive stage: cross-encoder reranking. This step—executed by a TinyBERT model—involves dynamic transformer inference for every (query, document) pair, significantly increasing average latency. In contrast, BM25 and TF-IDF rely solely on lightweight, CPU-based term-matching operations, resulting in far lower computational demands.

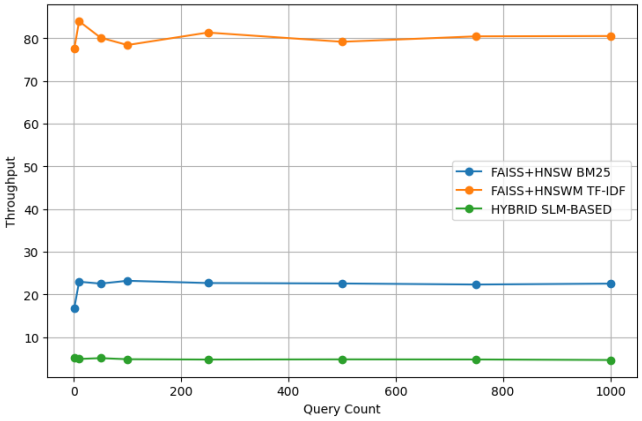


Fig. 4 Throughput after 1,000 FinQA Queries

The lower throughput of the SLM-based approach can be attributed to the limited scalability of the GPU-bound reranking stage, which processes query-document pairs serially or in small batches. On the other hand, Sparse retrievers utilize precomputed lexical scores and simple lookups, enabling higher query handling capacity per second.

TABLE II. Summary of Document Relevance Results

Retriever	Precision@5	Recall
Hybrid FAISS + HNSW BM25	0.012029	0.029724
Hybrid FAISS + HNSW TF-IDF	0.011286	0.030338
SLM-based Hybrid	**0.028000**	**0.102224**

Table II shows that the SLM-based hybrid significantly outperforms sparse hybrids in Precision@5 and Recall. While Hybrid FAISS + TF-IDF and FAISS + BM25 deliver modest

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Recall (~0.03), the SLM-based method achieves over 3x higher Recall (0.102224), demonstrating its superior ability to identify contextually relevant documents.

This improved retrieval quality is primarily attributed to the cross-encoder architecture, which allows deep semantic modeling. Unlike traditional sparse methods that depend on term frequency or keyword overlap, the SLM-based approach captures contextual alignment even without direct lexical matches. This SLM-based approach excels at identifying truly relevant documents and mitigating the common bias toward keyword-dense entries.

These findings underscore a critical trade-off in retrieval-augmented systems. While the SLM-based hybrid offers substantial improvements in document relevance, this comes with increased computational overhead, making it more suitable for quality-critical applications than real-time systems. Furthermore, it supports using SLM-based hybrid retrieval as an effective front-end component for RAG systems, especially in deployment contexts where compute budgets preclude full-scale LLM generation.

V. Conclusion & Recommendations

This study systematically evaluated hybrid retrieval strategies for Retrieval-Augmented Generation (RAG) systems in a resource-constrained environment. By analyzing both retrieval effectiveness (Precision@5, Recall) and system efficiency (latency, throughput, total execution time), the performance of classical sparse retrievers (BM25, TF-IDF), hybrid methods (FAISS+BM25, FAISS+TF-IDF), and semantic cross-encoder-based approach (TinyBERT) was assessed across small-scale (1-10 queries), medium-scale (up to 1000 queries) and large-scale (7,000 queries) experiments.

The results demonstrate that the hybrid retrieval methods consistently perform better than the classical approaches on all performance dimensions. Among these, the SLM-based Hybrid retriever achieved the highest retrieval quality, with Recall exceeding 0.10 and Precision@5 nearly tripling the performance of sparse counterparts. However, this gain comes with notable efficiency trade-offs, particularly in throughput and latency, with query times an order of magnitude slower than sparse baselines. The FAISS + TF-IDF hybrid method emerged as the most computationally efficient, having the lowest latency (~0.01s/query) and highest throughput (~71 QPS), although not as reliable with retrieval effectiveness. FAISS + BM25 offered a middle-ground alternative with stable latency (~0.045s) and moderate Recall and precision, making it suitable for use cases requiring performance balance.

Importantly, while hybrid retrieval has been explored in prior literature, evaluations of Small Language Model (SLM)-based reranking under constrained computational environments remain limited. This study addresses that gap by providing an empirical baseline for quality versus system-performance trade-offs. It underscores the importance of application-specific retriever selection—where quality-centric applications may benefit from SLM-based rerankers. At the same time, real-time or edge deployments may prioritize sparse or hybrid-optimized alternatives. Through the demonstrated results, we can conclude that SLM-based hybrid retrieval approaches can deliver high semantic relevance under realistic system constraints and at scale, validating their suitability for integration into real-world RAG pipelines operating over large open-domain corpora like OSCAR.

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